

Assignment 3A- global baseline estimate

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Approach

For this assignment, I applied a global baseline estimate to a dataset of user ratings for six historical TV dramas and predicted Participant-1's missing rating for the show “Shogun”.

1. Load Required Package

tidyverse is used for data manipulation (dplyr) and reshaping (tidyr)

```
library(tidyverse)
```

2. Read CSV from URL

Read the historical TV drama ratings dataset directly from GitHub into dataframe “ratings” & converts blank cells (missing ratings) to NA in R.

```
ratings <- read.csv( "https://raw.githubusercontent.com/KieraG2026/Global-Baseline-Estimate/refs/heads/main/historical\_tv\_drama\_ratings.csv",  
na.strings = c("", "NA")) head(ratings)
```

Check first few rows

Participant Bridgerton Peaky.Blinders Gilded.Age Underground Shogun
Warriors

1 Participant-1	2	3	5	4	NA	NA
2 Participant-2	4	NA	5	NA	NA	5
3 Participant-3	NA	5	1	5	4	NA
4 Participant-4	3	NA	5	5	4	NA
5 Participant-5	NA	NA	4	5	NA	5

3. Convert Wide Format to Long Format

Formats dataset into tidy style to use for modeling: Each row = one rating for one show.

```
ratings_long <- ratings %>% pivot_longer(  
  -Participant, # Keep Participant as identifier  
  names_to = "Show", # Column for TV show names  
  values_to = "Rating" # Column for ratings ) head(ratings_long)
```

Verify long format

	Participant	Show	Rating
	<chr>	<chr>	<int>
1	Participant-1	Bridgerton	2
2	Participant-1	Peaky.Blinders	3
3	Participant-1	Gilded.Age	5
4	Participant-1	Underground	4
5	Participant-1	Shogun	NA
6	Participant-1	Warriors	NA

4. Compute Global Mean

Average of all ratings, ignoring missing ratings. Starting point for Global Baseline Estimate. Consiering ratings are on a scale of 1 to 5, a 4.111 global mean shows users in this dataset tend to rate on the higher side.

```
global_mean <- mean(ratings_long$Rating, na.rm = TRUE) global_mean
```

Print global mean

```
[1] 4.111111
```

5. Compute User Average Ratings

Average rating for each participant (user). Greater than global means shows a user who rates generously.

```
user_avg <- ratings_long %>% group_by(Participant) %>%  
  summarize(user_mean = mean(Rating, na.rm = TRUE))
```

6. Compute User Bias (b_u)

How much each user deviates from the global mean. Deduct global mean (4.111) from user mean.

```
user_bias <- user_avg %>% mutate(b_u = user_mean - global_mean)  
user_bias
```

Check Participant-1 bias

Participant	user_mean	b_u
<chr>	<dbl>	<dbl>
1 Participant-1	3.5	-0.611
2 Participant-2	4.67	0.556
3 Participant-3	3.75	-0.361
4 Participant-4	4.25	0.139
5 Participant-5	4.67	0.556

7. Compute Show Average Ratings

Average rating for each TV show.

```
show_avg <- ratings_long %>% group_by>Show) %>% summarize(show_mean  
= mean(Rating, na.rm = TRUE))
```

8. Compute Show Bias (b_i)

How much each show deviates from the global mean (4.111). “Shogun” has a bias of –0.111 which is very close to neutral but still means it is less liked by users than some of the other shows rated.

```
show_bias <- show_avg %>% mutate(b_i = show_mean - global_mean)
show_bias
```

```
# Check bias for Shogun
```

```
# A tibble: 6 x 3
```

Show	show_mean	b_i
<chr>	<dbl>	<dbl>
1 Bridgerton	3	-1.111
2 Gilded.Age	4	-0.111
3 Peaky.Blinders	4	-0.111
4 Shogun	4	-0.111
5 Underground	4.75	0.639
6 Warriors	5	0.889

9. Apply Global Baseline Formula

Predicted rating for Participant-1 for “Shogun” would be ~3.39, lower than the global mean of 4.111. This is due to Participant-1's user bias of –0.611 (tendency to rate more critically) and “Shogun’s” show bias of –0.111 (less liked show among users).

```
bu <- user_bias$b_u[user_bias$Participant == "Participant-1"]
bi <- show_bias$b_i[show_bias$Show == "Shogun"]
prediction <- global_mean + bu + bi
prediction
```

```
# Print final prediction
```

```
[1] 3.388889
```


